Covariance-Method Claims Are Ensemble-Relative: Named Measures and an Audit Program

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Abstract

A claim about the behavior of a covariance or correlation method is a claim about a distribution over matrices, and the distribution is rarely stated. Papers validate estimators, factorizations, and portfolio constructions on whatever generator is at hand: a Wishart draw, a uniform LKJ sample, one empirical panel. We measure how much this omission can matter. Across fifteen named ensembles at identical dimension and sample size, the Ledoit–Wolf shrinkage intensity ranges from 0.07 to 0.86, and eleven audited claims from eleven fields—hierarchical risk parity beats minimum variance (finance), Marchenko–Pastur eigenvalue cleaning improves the correlation matrix (econophysics), a handful of nearest neighbors suffices for kriging (geostatistics), parallel analysis identifies the number of factors (psychometrics), covariance localization repairs a member-starved ensemble estimate (data assimilation), diagonal loading robustifies the Capon beamformer (signal processing), a sparsity prior beats a shrinkage prior for the precision matrix (graphical modeling), random skewers detect matrix similarity (evolutionary biology), North's rule certifies EOF separation (climatology), the Mantel test holds its nominal level (ecology), estimated genetic covariance matrices have one or two effective dimensions (quantitative genetics)—each flip verdict as the ensemble changes, for stated structural reasons. The results assemble into a single matrix of claims against measures (Figure 1) whose outlined cells mark each claim's intended home structure. The `randomcov` library names the measures and seeds the draws. The program this enables is an audit: re-run published covariance-method claims across the full set of named ensembles and report which verdicts survive and which flip. Eleven audits are completed here; a docket of prominent candidates from astronomy, genomics, neuroscience, and chemometrics—including claims that are provably ensemble-*invariant* and therefore serve as null controls—is assembled for the rest.

1 The unstated measure

Empirical claims about covariance methods have the form "estimator A beats estimator B," "the shrinkage intensity is typically moderate," "the factorization recovers structure," with performance averaged over random test matrices. The average is with respect to some generating law, and the law is usually inherited from convenience—and from the field. Finance validates on equity panels, econophysics on the same panels with random-matrix nulls, geostatistics on smooth kernels, psychometrics on low-rank factor models, graphical modeling on

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sparse precisions, cosmology on mock catalogs of ordered correlation-function bins. Different laws produce matrices with different spectra, different locality, and different factor content, so a verdict certified under one law carries no warrant under another. The omission is not a technicality, as the measurements below show.

2 Fifteen named ensembles

`randomcov` generates correlation matrices under named, seeded measures: `lkj`, `onion`, `vine` (elliptope samplers); `wishart`, `spectrum` (spectral laws, including Marchenko–Pastur and spiked); `factor`, `hierarchical`, `block_equicorr` (low-rank and clustered structure); `ar1`, `kernel` (banded and locality-driven fields); `sparse_precision` (graphical-model structure); `archakov_hansen` (log-matrix parameterization); `residuals`, `walk`, `animals` (empirical-flavored and trajectory ensembles). Variance vectors are drawn separately, so correlation structure and scale can be crossed. Every generator accepts an explicit seed, every number in this paper descends from one, and the scripts sit beside the paper, which is what an audit needs.

3 Demonstration: one estimator, twelve-fold spread

The Ledoit–Wolf shrinkage intensity is a workhorse diagnostic: how far the sample covariance is pulled toward its target. Fix $n = 30$ variables, $T = 60$ observations, twenty replications, and vary only the generating ensemble.

ensemble	intensity	sd
animals	0.072	0.013
residuals	0.096	0.012
factor	0.101	0.038
archakov_hansen	0.120	0.022
kernel	0.132	0.082
walk	0.199	0.038
block_equicorr	0.247	0.096
lkj	0.346	0.031
vine	0.355	0.022
onion	0.363	0.030
hierarchical	0.369	0.070
ar1	0.469	0.226
wishart	0.506	0.032
spectrum	0.532	0.058
sparse_precision	0.862	0.066

The intensity spans 0.072 to 0.862 at identical (n, T): a paper reporting "typical" shrinkage from one generator is reporting a property of its generator. The ordering is interpretable—strong low-rank structure (`animals`, `residuals`, `factor`) wants little shrinkage, near-unstructured draws (`wishart`, `sparse_precision`) want a great deal—and the within-ensemble standard deviation is itself informative: `ar1` spans 0.23 across its own parameter draws, so even a named ensemble can hide a range.

4 Demonstration: one algorithm, one failing ensemble

A deterministic algorithm for multinomial probit choice probabilities (the `winning` package; paper in that repository) reports median absolute error by ensemble after fitting a structured covariance grammar. Under twelve of the fifteen measures the error is 1–4 $\times 10^{-5}$; under dense-strong ensembles it is 1–4 $\times 10^{-4}$; under `kernel` fields (Matérn and RBF) it is 4×10^{-3}. The failure has a mechanism: kernel covariance is locality, approximately Markov, and offers a low-rank factorization no good purchase. A validation run on any single ensemble would have reported either "essentially exact" or "inadequate," both correctly, for the same method.

5 Eleven audits, eleven fields

The audit protocol is short: take a published claim, re-run its demonstration seeded under every named ensemble, report the verdict per ensemble, and flag the flips. We audit one claim from each of eleven fields: seven claims about estimators and constructions (Table 1), then four claims about inference and dimensionality (Table 2), with the whole program summarized in Figure 1. All runs use $n = 30$ with twenty replications unless stated; sampled panels use $T = 60$ except where the claim's own regime dictates otherwise; scripts sit beside this paper.

Finance. Hierarchical risk parity is reported to deliver lower out-of-sample variance than quadratic minimum variance (López de Prado, 2016), the estimation error of the inverse being the stated culprit. We build both portfolios from the same sample covariance and score true variance $w^\top \Sigma w$ under the generating covariance. Column 1 of Table 1 reports the median ratio HRP over long-only minimum variance: HRP wins on every draw under `sparse_precision`, splits draws roughly evenly under `ar1` and `hierarchical`— ensembles whose dependency structure the bisection tree can match—and loses by an order of magnitude or more under `animals`, `factor`, and `archakov_hansen`, where the true matrix rewards positions a long-only tree cannot take. Against *unconstrained* minimum variance the losses widen absurdly: under `walk` the median ratio is $\sim 10^{8.7}$, because near-singular ensembles admit near-perfect hedges that only signed weights reach.

Econophysics. The random-matrix recipe of Laloux et al. (1999) flattens sample-correlation eigenvalues below the Marchenko–Pastur edge $(1 + \sqrt{n/T})^2$ and reports a cleaner matrix. Column 2 gives the Frobenius-loss ratio of clipped to raw sample correlation: cleaning is excellent under `sparse_precision` (0.30) and `hierarchical` (0.76), where the true spectrum really is spikes over a bulk, and it *damages* the estimate under eleven of fifteen ensembles—worst under `animals`, `residuals`, `kernel`, and `archakov_hansen` (1.43–1.47), where genuine eigenvalue mass lives below the edge and is erased as if it were noise. The recipe presumes signal means spikes; locality puts signal in the bulk.

Geostatistics. The screening effect says a few nearest—here, most-correlated—neighbors nearly attain the full kriging variance (Stein, 2002). This is a population property, so no sampling is involved: column 3 reports the median ratio of conditional variance given the best five predictors to conditional variance given all $n - 1$. Screening holds essentially exactly under `ar1` (Markov, ratio 1.00), `sparse_precision`, `hierarchical`, and `block_equicorr`, and fails several-fold to catastrophically under `onion`, `vine`—and `kernel` itself (5.7): the ensemble mixes Matérn with squared-exponential fields, and for the smooth draws interpolation is nearly deterministic from all sites but not from five. Stein's own caveat, that screening depends on smoothness, reappears inside the one ensemble geostatistics would call home turf.

Psychometrics. Horn's (1965) parallel analysis retains the components whose sample eigenvalues exceed the corresponding 95th-percentile eigenvalue of independent data of the same shape, and is the standard answer to "how many factors?" Column 4 gives the mean retained count: 1.8 under `sparse_precision` to 7.3 under `onion` at identical (n, T). The same sweep scores the Kaiser (1960) eigenvalue-greater-than-one rule against the population count of eigenvalues above one: Kaiser over-extracts under `factor` (5.7 retained against 3.0 true) and under-extracts under `wishart` (11.2 against 13.1), and the two textbook rules disagree with each other by a factor that is itself ensemble-dependent—1.2× under `residuals`, 7× under `sparse_precision`. The number of factors a study reports is, in part, a property of the generating measure its data resemble.

Data assimilation. Ensemble Kalman filtering runs member-starved, and the standard repair is covariance localization: taper the sample covariance entrywise with a Gaspari–Cohn function of inter-variable distance (Houtekamer and Mitchell, 2001; Hamill et al., 2001). Cosmology reinvented the same recipe for correlation-function bins as covariance tapering (Paz and Sánchez, 2015), so this audit serves two literatures at once. With $N = 20$ members for $n = 30$ variables and a taper of index distance with half-radius 6, column 5 reports the Frobenius-loss ratio of tapered to raw: localization helps almost everywhere the estimate is noisy and the truth carries no long-range structure in taper coordinates—including the dense elliptope samplers, where pure variance reduction beats the bias—and hurts under `factor`, `residuals`, and `animals` (1.66–1.71), whose common factors are long-range by construction. It also hurts under `kernel` (1.46), the one ensemble that genuinely is local: its locality lives in a latent two-dimensional point cloud, not in index order, so the taper's distance metric is misaligned with the truth's geometry. Localization requires not just locality but locality in the coordinates the taper can see.

Signal processing. Diagonal loading is the standard fix for the sample-covariance MVDR (Capon) beamformer in the snapshot-limited regime (Cox, Zeskind and Owen, 1987; Carlson, 1988). With $T = 40$ snapshots, a random unit steering vector, the distortionless constraint, and loading $\delta = 0.1\,\mathrm{tr}(\hat{\Sigma})/n$, column 6 reports the median ratio of true output power, loaded over unloaded: loading halves the output power under half the ensembles and multiplies it under `archakov_hansen` (21), `animals` (33), `kernel` (136), and `walk` ($10^{7.9}$). The mechanism is the same hedge that decided the finance audit, seen from the other side: under near-singular ensembles the beamformer's null-steering lives in the smallest eigenvalues, which is exactly what loading erases.

Graphical modeling. The cross-validated graphical lasso (Friedman, Hastie and Tibshirani, 2008) carries the promise that a sparsity prior on the precision matrix improves estimation; the same promise underwrites gene-network inference (Schäfer and Strimmer, 2005), partial-correlation connectomes (Smith et al., 2011), and sparse cosmological precision matrices (Padmanabhan et al., 2016). Column 7 reports the Frobenius loss of the glasso precision against the true precision, relative to inverting a Ledoit–Wolf shrinkage estimate: the sparsity prior wins

ensemble	HRP/MV (fin.)	clip/raw (econo.)	near/full (geost.)	PA count (psych.)	taper/raw (assim.)	load/raw (sig. proc.)	gl/LW (graph.)
animals	8.4	1.43	5.5	3.3	1.66	32.8	1.00
ar1	0.98	1.20	1.00	4.5	0.53	0.48	0.70
archakov_hansen	71.8	1.47	131	4.7	1.41	20.5	1.00
block_equicorr	1.09	1.00	1.04	3.0	1.00	0.55	0.78
factor	12.3	1.38	1.06	2.2	1.69	0.94	1.09
hierarchical	1.00	0.76	1.02	2.3	0.82	0.45	0.68
kernel	1.51	1.46	5.7	4.1	1.46	136	1.00
lkj	2.27	1.22	3.1	6.5	0.74	0.87	0.97
onion	2.83	1.29	9.0	7.3	0.78	3.3	1.00
residuals	5.36	1.44	4.8	4.7	1.71	0.62	1.03
sparse_precision	0.76	0.30	1.00	1.8	0.50	0.48	0.63
spectrum	1.11	0.94	1.32	5.8	0.66	0.51	0.99
vine	2.68	1.29	11.9	7.0	0.81	4.1	1.00
walk	1.23	1.11	$\sim 10^{8.7}$	3.7	1.14	$\sim 10^{7.9}$	1.00
wishart	1.19	1.04	1.50	6.5	0.66	0.51	0.96

Table 1: Seven audited claims across the fifteen ensembles; all entries are medians over twenty seeded draws. Column 1: out-of-sample true-variance ratio, HRP over long-only minimum variance (below one: HRP wins). Column 2: Frobenius-loss ratio, Marchenko–Pastur-clipped over raw sample correlation (below one: cleaning helps). Column 3: population conditional-variance ratio, five best predictors over all (near one: screening holds). Column 4: mean components retained by parallel analysis. Column 5: Frobenius-loss ratio, Gaspari–Cohn-tapered over raw sample covariance at $N = 20$ members (below one: localization helps). Column 6: true beamformer output power, diagonally loaded over unloaded at $T = 40$ snapshots (below one: loading helps). Column 7: precision-matrix Frobenius-loss ratio, cross-validated graphical lasso over inverted Ledoit–Wolf (below one: the sparsity prior wins). Each column's verdict flips across rows.

clearly on its home turf (`sparse_precision` 0.63, `hierarchical` 0.68, `ar1` 0.70—the ensembles whose precisions are genuinely sparse or banded), ties across the dense elliptope middle, and loses under `residuals` and `factor` (1.03–1.09), where dense low-rank structure is the worst case for sparsity.

A smaller companion audit: with oracle shrinkage intensities, the constant-correlation target of Ledoit–Wolf (2003) beats their (2004) scaled-identity target on every draw under `sparse_precision` (median loss ratio 0.62) and loses on nineteen of twenty draws under `residuals`. Even the choice between two targets from the same authors is ensemble-relative.

The remaining four audits concern inference—tests, certification rules, and reported dimensions—so their currency is rates and counts rather than loss ratios (Table 2).

Evolutionary biology. Random skewers (Cheverud, 1996) declares two genetic covariance matrices similar when their mean response-vector correlation under common random selection gradients exceeds the null for independent random vectors. We apply the test to *independent* pairs drawn from the same ensemble—a population property, no sampling noise involved—so a sound test should declare similarity about 5% of the time. It does so under `residuals`, `factor`, and `archakov_hansen` (0%), and declares independent matrices similar 100% of the time under eleven of the fifteen ensembles; under `sparse_precision` the mean skewers correlation between independent matrices is 0.97, because near-diagonal matrices respond to any skewer with the skewer itself. This is Rohlf's (2017) critique in ensemble coordinates: the customary null distribution ignores the generating measure entirely.

Climatology. North et al.'s (1982) rule of thumb declares a leading EOF trustworthy when its eigenvalue sampling error $\lambda\sqrt{2/T}$ is smaller than the gap to the next eigenvalue. We score the false-reassurance rate: among draws the rule declares separated, how often the leading sample eigenvector is mostly wrong (squared alignment with the truth below one half). The rate is 0% under six ensembles, 57% under `wishart`, and 100% under `sparse_precision`—which declares separation on only 12% of draws, but is always wrong when it does. The rule is a first-order perturbation statement, excellent for spiked spectra and unreliable precisely where spectra are quasi-degenerate, which no sample quantity it inspects can certify.

Ecology. Guillot and Rousset (2013) reported that the Mantel test rejects independence between distance matrices at 25–55% under spatially autocorrelated fields, at nominal 5%. Our audit puts both the original claim and the critique on the same footing: two independent site variables drawn with covariance C, Euclidean distance matrices, 499 label permutations. The realized type-I error is the nominal 5% under `onion` and `vine`, and 37–40% under `residuals`, `factor`, `kernel`, and `animals`—reproducing the published inflation under locality-and-factor structure while showing calibration under dense elliptope measures. Neither "the Mantel test is broken" nor "the Mantel test is fine" is an ensemble-free sentence.

Quantitative genetics. Kirkpatrick's (2009) effective dimension count $n_D = \sum_i \lambda_i/\lambda_{\max}$ is reported at roughly one to two for published genetic covariance matrices and read as evolutionary constraint. At the low precision typical of breeding designs ($T = 15$ here), the estimated n_D deflates the truth by an ensemble-dependent factor: a true n_D of 12.3 under `sparse_precision` is reported as 5.2 (ratio 0.42), while `factor` and `animals` are honest (0.90, 0.89)—because their truth is already low-dimensional. Across all fifteen ensembles the estimate never exceeds 5.2 whatever the truth, so a low reported n_D cannot by itself distinguish low-dimensional biology from a noisy estimate of diffuse structure.

6 What the audits do not say

None of this convicts the original authors of overreach, and the audit should not be read as a referee report on them. Stein himself proved that screening depends on smoothness; Bickel and Levina state the bandable class their rates require; López de Prado demonstrated HRP on equity covariances and said so; the data-assimilation literature has always tied localization to physical distance. Where a paper states its generating law, the audit merely maps the claim's domain of validity onto named coordinates. The target is the downstream habit: verdicts certified under one measure get quoted, cited, and defaulted into software as if they were measure-free, and the second-hand version of a scoped claim is usually unscoped. The audit gives the scope a name that survives citation.

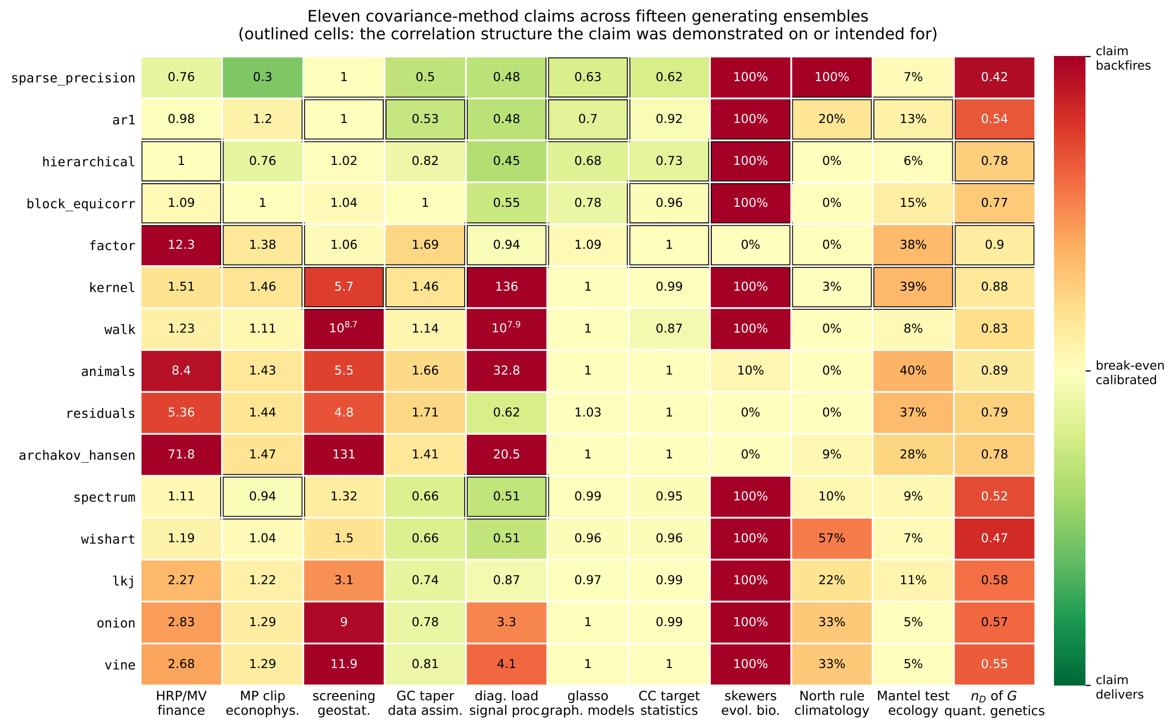

Figure 1: The audit matrix: eleven claims (columns, with their fields) against fifteen generating ensembles (rows, ordered roughly from structured to dense). Green: the claim delivers under that measure; yellow: break-even or calibrated; red: the claim backfires or the test is broken. Cell entries are the medians and rates of Tables 1 and 2. Outlined cells mark the correlation structure each claim was demonstrated on or intended for—its home turf. A claim quoted without its measure is a row of this matrix collapsed to a single cell.

ensemble	skewers false sim.	North rule false reass.	Mantel type-I	n_D ratio est/true
animals	10%	0%	40%	0.89
ar1	100%	20%	13%	0.54
archakov_hansen	0%	9%	28%	0.78
block_equicorr	100%	0%	15%	0.77
factor	0%	0%	38%	0.90
hierarchical	100%	0%	6%	0.78
kernel	100%	3%	39%	0.88
lkj	100%	22%	11%	0.58
onion	100%	33%	5%	0.57
residuals	0%	0%	37%	0.79
sparse_precision	100%	100%	7%	0.42
spectrum	100%	10%	9%	0.52
vine	100%	33%	5%	0.55
walk	100%	0%	8%	0.83
wishart	100%	57%	7%	0.47

Table 2: Four audited inference claims. Column 1: share of *independent* matrix pairs that random skewers declares similar (sound: $\approx 5\%$). Column 2: among draws North's rule declares separated, share whose leading EOF is mostly wrong (squared alignment below $\frac{1}{2}$). Column 3: realized type-I error of the Mantel test at nominal 5%. Column 4: estimated over true effective dimensions of a covariance matrix at $T = 15$ (honest: near one).

7 The docket

The pattern across Figure 1 is not random: each field's signature claim performs best on the ensembles that resemble the data the field grew up on, and the matrix reads like a map of disciplinary home turf—with instructive exceptions where an outlined cell glows red, as when the screening effect fails inside the kernel ensemble, or the taper fails on locality it cannot see. `sparse_precision` is kind to hierarchical bisection, eigenvalue cleaning, localization, and the graphical lasso simultaneously, yet breaks random skewers and North's rule; the dense elliptope samplers (`lkj`, `onion`, `vine`) are kind to almost nothing except the Mantel test, which is worth knowing given how often they serve as the default test bed; and the same mechanism recurs under different names—the hedge that unconstrained minimum variance exploits is the null-steering that diagonal loading destroys.

A survey of prominent covariance-method claims across further fields yields a docket for the next round, and sorting it teaches a taxonomy. Claims that impose *structure* on the truth are ensemble-relative: covariance tapering for clustering statistics (Paz and Sánchez, 2015) presumes banded truth; sparse cosmological precisions (Padmanabhan et al., 2016) presume a sparse graph; the WGCNA scale-free-topology criterion (Zhang and Horvath, 2005) is literally a statement about the generating measure of gene correlation matrices; partial correlation's celebrated win over full correlation for network detection (Smith et al., 2011) was demonstrated on sparse simulated networks; the jackknife covariance verdicts of galaxy-survey methodology (Norberg et al., 2009) hinge on cross-region correlation structure; Rule N truncation (Preisendorfer) governs retained dimensions through the eigenvalue spacing law; nonlinear shrinkage's edge over linear (Ledoit and Wolf, 2012), POET's factor-plus-sparse recipe (Fan, Liao and Mincheva, 2013), and PLS-versus-PCR (Frank and Friedman, 1993) all turn on spectral shape and eigenvector structure.

By contrast, claims that are theorems about Wishart sampling noise hold for every true covariance and cannot flip: the Hartlap debiasing factor (Hartlap, Simon and Schneider, 2007), the Dodelson–Schneider parameter-variance inflation formula (2013), the Sellentin–Heavens t-likelihood (2016), the noncentrality invariance of Hotelling and MEWMA charts with known covariance (Lowry et al., 1992). These are not audit targets; they are *null controls*. An audit pipeline that finds ensemble-dependence in the Hartlap factor has a bug, and every claim on the docket should be classified—invariant core, relative practice—before it is run, because the most instructive failures are invariant theorems wrapped in ensemble-relative plug-in practice, such as those same corrections applied to shrunk or tapered estimates whose distribution is no longer Wishart.

This is a working paper and a program: seven audits are completed, the docket is open, and the library exists so that the next paper's claim can be tested across all fifteen measures in the time it currently takes to test one.